

Student Performance Project

Dataset Information

Additional Information

This data approach student achievement in secondary education of two Portuguese schools. The data attributes include student grades, demographic, social and school related features) and it was collected by using school reports and questionnaires. Two datasets are provided regarding the performance in two distinct subjects: Mathematics (mat) and Portuguese language (por). In [Cortez and Silva, 2008], the two datasets were modeled under binary/five-level classification and regression tasks. Important note: the target attribute G3 has a strong correlation with attributes G2 and G1. This occurs because G3 is the final year grade (issued at the 3rd period), while G1 and G2 correspond to the 1st and 2nd period grades. It is more difficult to predict G3 without G2 and G1, but such prediction is much more useful (see paper source for more details).

Feature Details

Variable Name	Role	Type	Demographic	Description	Units	Missing Values
school	Feature	Categorical		student's school (binary: 'GP' - Gabriel Pereira or 'MS' - Mousinho da Silveira)		no
sex	Feature	Binary	Sex	student's sex (binary: 'F' - female or 'M' - male)		no
age	Feature	Integer	Age	student's age (numeric: from 15 to 22)		no
address	Feature	Categorical		student's home address type (binary: 'U' - urban or 'R' - rural)		no
famsize	Feature	Categorical	Other	family size (binary: 'LE3' - less or equal to 3 or 'GT3' - greater than 3)		no
Pstatus	Feature	Categorical	Other	parent's cohabitation status (binary: 'T' - living together or 'A' - apart)		no
Medu	Feature	Integer	Education Level	mother's education (numeric: 0 - none, 1 - primary education (4th grade), 2 - 5th to 9th grade, 3 - secondary education or 4 - higher education)		no
Fedu	Feature	Integer	Education Level	father's education (numeric: 0 - none, 1 - primary education (4th grade), 2 - 5th to 9th grade, 3 - secondary education or 4 - higher education)		no
Mjob	Feature	Categorical	Occupation	mother's job (nominal: 'teacher', 'health' care)		no

				related, civil 'services' (e.g. administrative or police), 'at_home' or 'other')		
Fjob	Feature	Categorical	Occupation	father's job (nominal: 'teacher', 'health' care related, civil 'services' (e.g. administrative or police), 'at_home' or 'other')		no
reason	Feature	Categorical		reason to choose this school (nominal: close to 'home', school 'reputation', 'course' preference or 'other')		no
guardian	Feature	Categorical		student's guardian (nominal: 'mother', 'father' or 'other')		no
traveltime	Feature	Integer		home to school travel time (numeric: 1 - <15 min., 2 - 15 to 30 min., 3 - 30 min. to 1 hour, or 4 - >1 hour)		no
studytime	Feature	Integer		weekly study time (numeric: 1 - <2 hours, 2 - 2 to 5 hours, 3 - 5 to 10 hours, or 4 - >10 hours)		no
failures	Feature	Integer		number of past class failures (numeric: n if 1<=n<3, else 4)		no
schoolsup	Feature	Binary		extra educational support (binary: yes or no)		no
famsup	Feature	Binary		family educational support (binary: yes or no)		no
paid	Feature	Binary		extra paid classes within the course subject (Math or Portuguese) (binary: yes or no)		no
activities	Feature	Binary		extra-curricular activities (binary: yes or no)		no
nursery	Feature	Binary		attended nursery school (binary: yes or no)		no
higher	Feature	Binary		wants to take higher education (binary: yes or no)		no
internet	Feature	Binary		Internet access at home (binary: yes or no)		no
romantic	Feature	Binary		with a romantic relationship (binary: yes or no)		no
famrel	Feature	Integer		quality of family relationships (numeric: from 1 - very bad to 5 - excellent)		no
freetime	Feature	Integer		free time after school (numeric: from 1 - very low to 5 - very high)		no

goout	Feature	Integer		going out with friends (numeric: from 1 - very low to 5 - very high)		no
Dalc	Feature	Integer		workday alcohol consumption (numeric: from 1 - very low to 5 - very high)		no
Walc	Feature	Integer		weekend alcohol consumption (numeric: from 1 - very low to 5 - very high)		no
health	Feature	Integer		current health status (numeric: from 1 - very bad to 5 - very good)		no
absences	Feature	Integer		number of school absences (numeric: from 0 to 93)		no
G1	Target	Categorical		first period grade (numeric: from 0 to 20)		no
G2	Target	Categorical		second period grade (numeric: from 0 to 20)		no
G3	Target	Integer		final grade (numeric: from 0 to 20, output target)		no

Additional Variable Information

Attributes for both student-mat.csv (Math course) and student-por.csv (Portuguese language course) datasets:

- 1 school - student's school (binary: 'GP' - Gabriel Pereira or 'MS' - Mousinho da Silveira)
- 2 sex - student's sex (binary: 'F' - female or 'M' - male)
- 3 age - student's age (numeric: from 15 to 22)
- 4 address - student's home address type (binary: 'U' - urban or 'R' - rural)
- 5 famsize - family size (binary: 'LE3' - less or equal to 3 or 'GT3' - greater than 3)
- 6 Pstatus - parent's cohabitation status (binary: 'T' - living together or 'A' - apart)
- 7 Medu - mother's education (numeric: 0 - none, 1 - primary education (4th grade), 2 - 5th to 9th grade, 3 - secondary education or 4 - higher education)
- 8 Fedu - father's education (numeric: 0 - none, 1 - primary education (4th grade), 2 - 5th to 9th grade, 3 - secondary education or 4 - higher education)
- 9 Mjob - mother's job (nominal: 'teacher', 'health' care related, civil 'services' (e.g. administrative or police), 'at_home' or 'other')
- 10 Fjob - father's job (nominal: 'teacher', 'health' care related, civil 'services' (e.g. administrative or police), 'at_home' or 'other')

11 reason - reason to choose this school (nominal: close to 'home', school 'reputation', 'course' preference or 'other')

12 guardian - student's guardian (nominal: 'mother', 'father' or 'other')

13 traveltime - home to school travel time (numeric: 1 - <15 min., 2 - 15 to 30 min., 3 - 30 min. to 1 hour, or 4 - >1 hour)

14 studytime - weekly study time (numeric: 1 - <2 hours, 2 - 2 to 5 hours, 3 - 5 to 10 hours, or 4 - >10 hours)

15 failures - number of past class failures (numeric: n if $1 \leq n < 3$, else 4)

16 schoolsup - extra educational support (binary: yes or no)

17 famsup - family educational support (binary: yes or no)

18 paid - extra paid classes within the course subject (Math or Portuguese) (binary: yes or no)

19 activities - extra-curricular activities (binary: yes or no)

20 nursery - attended nursery school (binary: yes or no)

21 higher - wants to take higher education (binary: yes or no)

22 internet - Internet access at home (binary: yes or no)

23 romantic - with a romantic relationship (binary: yes or no)

24 famrel - quality of family relationships (numeric: from 1 - very bad to 5 - excellent)

25 freetime - free time after school (numeric: from 1 - very low to 5 - very high)

26 goout - going out with friends (numeric: from 1 - very low to 5 - very high)

27 Dalc - workday alcohol consumption (numeric: from 1 - very low to 5 - very high)

28 Walc - weekend alcohol consumption (numeric: from 1 - very low to 5 - very high)

29 health - current health status (numeric: from 1 - very bad to 5 - very good)

30 absences - number of school absences (numeric: from 0 to 93) # these grades are related with the course subject, Math or Portuguese:

31 G1 - first period grade (numeric: from 0 to 20)

31 G2 - second period grade (numeric: from 0 to 20)

32 G3 - final grade (numeric: from 0 to 20, output target)